

HisabDo – AI Content Recommendation System

Documentation

The AI Content Recommendation System is designed to help HisabDo users find relevant FAQs, guides, and other useful content based on their questions, interests, behavior, language, and previously viewed content.

1. Introduction

The system analyzes user questions and interactions and recommends content that is most useful to the user. The goal is to reduce the time users spend searching for information and improve content discovery.

2. Objectives

- Provide personalized content suggestions.
- Recommend content related to the user's current question.
- Use user behavior to improve recommendations.
- Recommend relevant FAQs and guides.
- Show recently viewed content.
- Support language-based recommendations.
- Use AI to understand the meaning and intent of user questions.
- Improve user engagement with HisabDo content.

3. Recommendation Types

Personalized Content Recommendations: Suggest content based on a user's previous searches, views, and interests.

Related Content Recommendations: Suggest FAQs, guides, or articles related to the content currently being viewed.

User Behavior-Based Recommendations: Use searches, clicks, views, and other interactions to understand user interests.

Smart FAQ and Guide Recommendations: Understand a user's question and recommend the most relevant FAQ or guide.

Recently Viewed Content: Show content that the user has recently opened.

Language-Based Recommendations: Prefer content written in the same language as the user's question.

AI-Powered Content Discovery: Use AI to understand the meaning of a question and find relevant content, even when exact keywords are different.

4. Required Data

FAQ and Content Data

Field	Description
FAQ ID	Unique identifier for the FAQ
Question	Question that the FAQ answers
Answer	Answer to the question
Category	For example: Budgeting, Expenses, Savings
Keywords	Important words related to the content
Related Content	Other FAQs or guides related to this content
Language	English, Urdu, or another supported language
Content Type	FAQ, Guide, Article, etc.

User Activity Data

- User searches
- FAQs or guides viewed
- Recommendations clicked
- Recently viewed content
- Interaction time
- Preferred language

5. System Workflow

USER → Question / Activity → Language Detection → Intent Detection → Content Retrieval → Relevance Ranking
→ Recommended Content → USER

The system receives a question or user activity, identifies the language and topic, finds relevant content, ranks the results, and presents the best recommendations.

6. Architecture Components

Frontend: Collects user questions and activity and displays recommendations.

Recommendation Service: Handles intent detection, language detection, content matching, and ranking.

FAQ / Content Database: Stores FAQs, guides, articles, categories, keywords, and language information.

User Activity Database: Stores searches, views, clicks, and other interaction data.

Embedding / Vector Search: An advanced option that finds content based on meaning and semantic similarity.

7. Example Use Case

User question: “How can I save more money every month?”

- Detect the user's language.
- Identify the topic as Savings / Budgeting.
- Search the FAQ and content dataset.
- Find relevant content.
- Rank the results.
- Display the best recommendations.

Possible recommendations: Monthly Savings Guide, Monthly Budget Guide, How to Reduce Expenses.

8. AI-Powered Content Discovery

Keyword matching only looks for similar words. AI can understand similar meanings. For example, the question “I want to control where my money goes every month” can be understood as a budgeting-related question even though the word “budget” is not used.

For an advanced implementation, embeddings and vector search can be used to find content with similar meaning.

9. Recommendation Ranking

Recommendations can be ranked using several signals:

- Similarity to the user's question.
- User interests and previous activity.
- Content category.
- Language match.
- Recent user activity.

10. Testing Requirements

Test Question	Expected Topic	Expected Recommendation
How can I manage my budget?	Budgeting	Budget Guide
How can I reduce expenses?	Expenses	Expense Guide
How can I save money?	Savings	Savings Guide
Mera budget kaise manage hoga?	Budgeting	Urdu Budget FAQ
How do I track my expenses?	Expenses	Expense Tracking FAQ

11. Evaluation Criteria

- **Relevance:** Is the recommended content related to the user's question?
- **Accuracy:** Did the system identify the correct topic or intent?
- **Language:** Is the content provided in the correct language?

- **Usefulness:** Does the recommendation help the user?
- **Ranking:** Is the most relevant content shown first?
- **Diversity:** Do recommendations provide useful alternatives instead of repeating the same content?

12. Initial MVP

The first version should focus on the FAQ dataset, content categories, question and intent detection, related FAQ recommendations, language detection, recently viewed content, basic recommendation ranking, test cases, and evaluation.

13. Future Enhancements

- AI embeddings
- Vector database and semantic search
- More supported languages
- Advanced personalization
- Recommendation feedback
- A/B testing
- Real-time recommendations
- Continuous improvement using user interactions

14. Project Deliverables

- FAQ and content dataset
- Content categories
- Recommendation scenarios and rules
- Test questions
- Expected recommendations
- AI evaluation results
- Recommendation workflow
- Technical documentation

15. Final Workflow

FAQ Dataset → User Question → Language Detection → Intent Detection → Find Relevant Content → Rank Results → Recommend Content → Evaluate Results

The main purpose is to give the right content to the right user at the right time.